

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B.Tech. Examination****May 2015****IT 201.01 Database Management System****Date: 15.05.2015, Friday****Time: 10:00 a.m. To 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I**Q - 1 Answer the following questions. [07]****(a) The number of entity sets linked through a single relationship set is called _____. [01]**

- | | |
|------------------------------|------------------------|
| A Degree of Relationship set | B Degree of Entity set |
| C Degree of Attribute set | D All of the above |

(b) In E-R diagram generalization process is represented by [01]

- | | |
|-------------|--------------------|
| A Ellipse | B Dashed Ellipse |
| C Rectangle | D Reverse Triangle |

(c) In relational algebra, Cartesian product is _____operator. [01]

- | | |
|----------------------|---------------------|
| A An Unary Operator | B A Binary Operator |
| C A Ternary Operator | D All of the above |

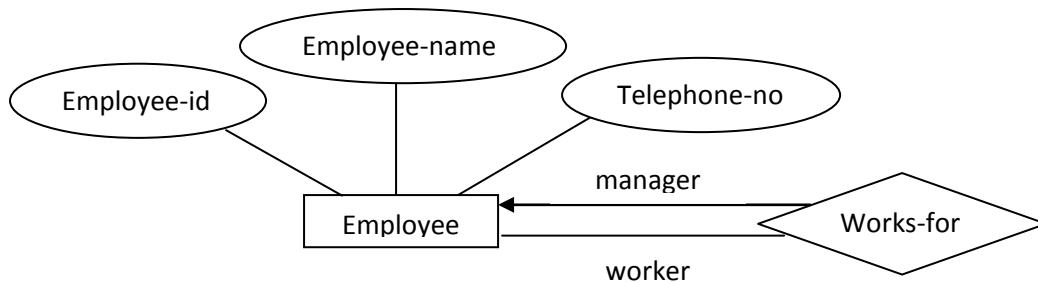
(d) What do you mean by ternary Relationship in E-R Diagram? [01]**(e) What is the difference between procedural query language and non-procedural / declarative query language? [01]****(f) Draw three level (abstract level) architecture of database management system. [02]****Q - 2 (a) Consider the following relational database, [06]**

works (person_name, company_name, salary);
 lives (person_name, street, city);
 located in (company_name, city);
 managers (person_name, manager_name);

Give an expression in the **Relational Algebra** for each request.

- 1) Find the names of the persons who work for company 'FBC'.
- 2) List out the person name with their street, city and salary.
- 3) Find out the average salary for each company.
- 4) Rename table 'works' to 'works1'.

(b) Justify the following statement according to the given example: [04]
 "Entity sets of a relationship need not be distinct".



- (c) Explain the concepts of specialization and generalization in detail with example. [04]

OR

- Q - 2 (a) Explain the following relational algebra operator with example. [06]

1) Project 2) Natural Join 3) Cartesian Product

- (b) Define the following terms related to the different types of attributes in E-R model : [04]
Composite attribute, Multi-valued attribute, Derived attribute and Domain.

- (c) Explain the Strong Entity Sets and Weak Entity Sets in E-R model with suitable example. [04]

- Q - 3 (a) A Relation $R = (A, B, C)$ with $F = \{A \rightarrow B, B \rightarrow C\}$ dependency is decomposed into $R_1 = (A, B)$ & $R_2 = (B, C)$. Find out the given decomposition is lossless or not. [04]

- (b) Find out the canonical cover / irreducible set of FD for following: [06]

$R = (A, B, C)$

$F = \{A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow C\}$

OR

- (b) Consider the following relation with given functional dependency. [06]

$R = (A, B, C, G, H, I)$

$F = \{A \rightarrow B, A \rightarrow C, CG \rightarrow H, CG \rightarrow I, B \rightarrow H\}$

Apply attribute closure $(AG)^+$ and check whether AG is the key attribute for above relation or not?

- (c) Differentiate between the physical data independence and logical data independence. [04]

SECTION – II

- Q - 4 Answer the following questions. [07]

- (a) In a functional dependency $X \rightarrow Y$, if Y is functionally dependent on X, but not on X's proper subsets, then we would call the functional dependency as, [01]

A Full Functional Dependency	B Partial Functional Dependency
C Multi-valued Functional Dependency	D None of the above

- (b) Which of the following would be the appropriate term for a key which consists of more than one attribute? [01]

A Composite Key	B Candidate Key
C Primary Key	D None of the above

- (c) Two sets of Functional Dependencies F and G are considered equal if $F^+ = G^+$. [01]

A True	B False
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- | | | |
|------------|---|------|
| (d) | Define the term “Transaction” related to database management system. | [01] |
| (e) | Define “Starvation” with suitable example. | [01] |
| (f) | Consider the following transaction involving two bank accounts x and y, | [02] |

```
read(x);
x:= x - 50;
write(x);
read(y);
y:= y + 50;
write(y);
```

The constraint that the sum of the accounts x and y should remain constant is that of,

- | | | | |
|---|-----------|---|-------------|
| A | Atomicity | B | Consistency |
| C | Isolation | D | Durability |

- Q - 5 (a)** Consider the following sequence of actions, and tell Whether, **[04]**
1. It is a conflict-serializable schedule or not,
 2. It is a view-serializable schedule or not

S: r1(x), w2(x), r3(x), r1(y), r4(z), w2(y), r1(v), w3(v), r4(v), w4(y), w5(y), w5(z)

- (b)** List the property of transaction. Draw and explain the state diagram of transaction. **[06]**

OR

- Q - 5 (a)** Consider the following sequence of actions, and tell Whether, **[04]**
1. It is a conflict-serializable schedule or not,
 2. It is a view-serializable schedule or not

S: r3(B), w1(A), w3(B), r1(B), r2(A), w3(A), w2(A)

- (b)** Explain the two-phase locking protocol of concurrency control mechanism in detail. **[06]**

- Q - 5 (c)** Explain the wait-die and wound-wait protocols for dead-lock prevention techniques. **[04]**

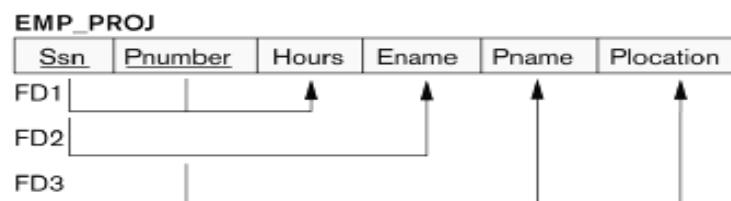
- Q - 6 (a)** List out the rules for 1st Normal form, 2nd Normal form and 3rd Normal form of normalization process. **[06]**

- (b)** Draw and explain the basic steps in query processing. [04]

- (c)** Explain the concept of checkpoint with example in recovery management system of database. **[04]**

OR

- Q - 6 (a)** Consider the following schema where {Ssn, Pnumber} is the Composite Key. [06]



Convert the above schema into 2nd Normal form.

- (b)** Explain the shadow-copy method of transaction in detail. **[04]**

- (c) Explain log-based recovery technique in detail. [04]

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